[Edit: 1/9 - Course will be online until the end of January. Office hours will be on Zoom at https://washington.zoom.us/my/pisan]

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q (425) 352-3741

Class: Tue/Thu 11:00-1:00pm. First week on Zoom, rest in UW1-102

Office Hours: Monday 1-2pm and Thursday 1-2pm. Signup via Canvas Calendar

Class Discord: invite link

Course Description

Principal ideas and developments in artificial intelligence, such as problem solving, knowledge representation, search, reasoning under uncertainty, learning, and natural language processing.

Course Learning Goals

ABET Outcomes:

1. an ability to apply knowledge of mathematics, science, and engineering;

2. an ability to design a system, component, or process to meet desired needs within

3. realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability;

4. an ability to use the techniques, skills, and modern engineering tools;

5. knowledge of contemporary issues;

6. a recognition of the need for, and an ability to engage in lifelong learning;

7. an ability to identify, formulate, and solve engineering problems; and

8. an ability to design and conduct experiments, as well as to analyze and interpret data

At the end of this course, students will be able to:

1. Describe what artificial intelligence (AI) means and how machines can process information intelligently;

2. Identify the different fields that comprise AI as well as techniques and heuristics for each area;

3. Create a formal representation of a complex problem; and

4. Write programs to solve complex problems using AI methods in areas such as search, reasoning, natural language processing, games, and robotics.

Textbooks

[Reference] Russell, Stuart J., and Norvig, Peter. *Artificial Intelligence : a Modern Approach*. 3rd edition, Pearson, 2015. http://aima.cs.berkeley.edu/ and https://www.amazon.com/Artificial-Intelligence-Approach-Stuart-Russell/dp/9332543518

Grading

- Exercises: 15%

- Projects: 45%

- Midterm: 20%

- Final: 20%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Exercises: You can miss one exercise without penalty. Exercises will be in the form of quizzes or other work building on what we are learning in class.

Projects: You can work individually or in pairs. When working as a pair, all substantial work must be performed with both students present (virtually). You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, post a brief explanation to "Using Extension" assignment.

There will be 4 projects of equal weight:

- Search (Week-2)

- Multi-Agent Search (Week 5)

- Reinforcement Learning (Week 7)

- Natural Language (Week 9)

Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct: http://www.washington.edu/cssc/for-students/student-code-of-conduct/ I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Midterm: The midterm will be in Week-6 during regular class hour. For the midterm, you are allowed a cheat sheet (letter-size, double-sided handwritten or typed)

Final: The final will be held during finals week. It will be similar to the midterm in terms of structure and you will once again have a cheat sheet.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the School of STEM Course Policies, which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources
- Religious Accommodations

- Respect for Diversity

- Student Support Services

- Surviving Sexual and Relationship Violence

- Wonder How to Address Faculty?